Personal Finance Intelligence Dashboard

A Power BI-Based Interactive Financial Analytics System

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Contents

1 Problem Statement **2**
1.1 Issues Identified 2
1.2 Objective 2

2 Solution and Features **2**
2.1 Data Architecture 2
2.2 Dashboard Pages 3
2.3 Key DAX Measures 3

3 Screenshots **4**
3.1 Dashboard 1 — Financial Overview 4
3.2 Dashboard 2 — Expense Intelligence 4
3.3 Dashboard 3 — Smart Insights 5
3.4 Dashboard 4 — Savings Forecast 5
3.5 Dashboard 5 — Financial Health Score 6

4 Technology Stack **6**

5 Unique Points **6**

6 Future Improvements **7**

7 Conclusion **8**

1. Problem Statement

Tracking personal finances is a challenge that most individuals face. Although banking apps and spreadsheets are widely available, they do not provide a clear picture of where money is going or whether savings goals are being met. Data remains scattered across different platforms and there is no way to interact with it or explore different financial scenarios.

1.1 Issues Identified

- **No centralized view:** Income and expense data exists across different platforms. There is no single report that consolidates everything together.
- **No budget tracking:** Monthly budgets are set but rarely reviewed to verify whether actual spending remained within the defined limits.
- **No future planning:** Most tools only display historical data. There is no mechanism to simulate the effect of changes in spending behaviour.
- **No overall score:** There is no straightforward way to assess financial health based on multiple factors at the same time.
- **Static reports:** Spreadsheets and PDFs cannot be filtered, drilled into, or explored interactively.

1.2 Objective

The objective of this project was to build an interactive personal finance dashboard in Microsoft Power BI that consolidates income, expenses, budget, and savings data. The aim was to ensure that users can understand their financial situation, identify problem areas, and plan ahead, all within a single platform.

2. Solution and Features

The project is a 5-page Power BI dashboard built on a Star Schema data model. The dataset covers 12 months of the year 2024 and contains over 1,000 transaction records distributed across 10 spending categories.

2.1 Data Architecture

The data is structured as a Star Schema with one central fact table and supporting dimension tables.

Table	Description
Transactions	Main fact table containing income and expense rows, transaction date, category, payment method, and a weekend flag
Calendar	Date dimension table with Month, MonthName, MonthYear, Quarter, Year, DayName, and IsWeekend columns
Categories	Bridge table linking Transactions to the Budget table across 10 spending categories
Budget	Monthly budget targets set for each spending category

2.2 Dashboard Pages

1. **Financial Overview:** Displays four KPI cards for Total Income, Total Expenses, Net Savings, and Savings Rate. Also includes an Income vs. Expenses trend line chart, a Net Savings area chart, and a monthly expense column chart.

2. **Expense Intelligence:** Analyses spending by category using a donut chart, a ranked bar chart, and a stacked column chart by month. Also includes a day-of-week spending analysis and a budget vs. actual table with colour-coded status.

3. **Smart Insights:** Displays four text cards that automatically generate sentences based on the data, covering weekend spending patterns, month-on-month changes, savings status, and the top spending category. A gauge and a MoM change chart are also included.

4. **Savings Forecast:** Uses a What-If slider to simulate different expense scenarios and shows the projected impact on savings. Also includes a Predicted Savings card and a 3-Month Moving Average chart.

5. **Financial Health Score:** Calculates a score out of 100 based on Savings Rate, Budget Control, and Spending Stability. Displayed as a gauge with supporting score cards and a breakdown bar chart.

2.3 Key DAX Measures

Measure	Logic
Total Income	Sums the Income_Amount column across all transactions
Total Expenses	Sums the Expense_Amount column
Net Savings	Total Income minus Total Expenses
Savings Rate %	Net Savings divided by Total Income, multiplied by 100
3M Moving Average	Average of expenses across the current and previous two months, recalculated each month using FILTER on the Transactions table
Scenario Savings	Income minus adjusted expenses based on the What-If slider percentage
Financial Health Score	A composite of three sub-scores: Savings (up to 40 pts), Budget (up to 35 pts), and Stability (up to 25 pts)
Budget Status	Classifies budget usage as On Track, Warning, or Overspent based on the spending percentage against budget
Insight Weekend	A text measure that generates a sentence comparing per-day weekend and weekday spending

3. Screenshots

3.1 Dashboard 1 — Financial Overview

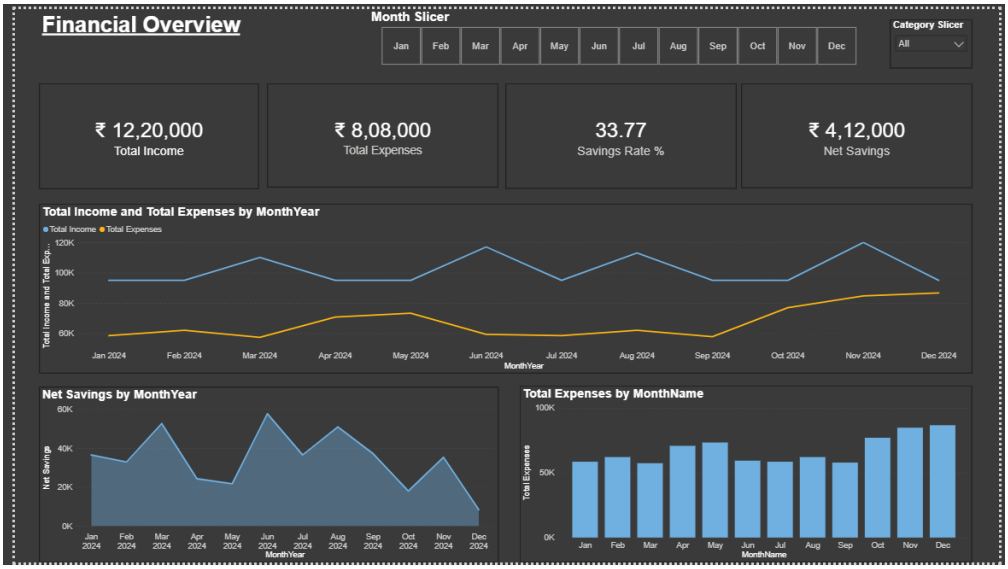

Figure 1: *Financial Overview showing KPI cards, Income vs. Expenses trend chart, Net Savings area chart, and monthly column chart*

The total annual income was Rs. 12,20,000 and total expenses stood at Rs. 8,08,000, resulting in net savings of Rs. 4,12,000 with a savings rate of 33.77%. The line chart confirms that income remained above expenses throughout the year.

3.2 Dashboard 2 — Expense Intelligence

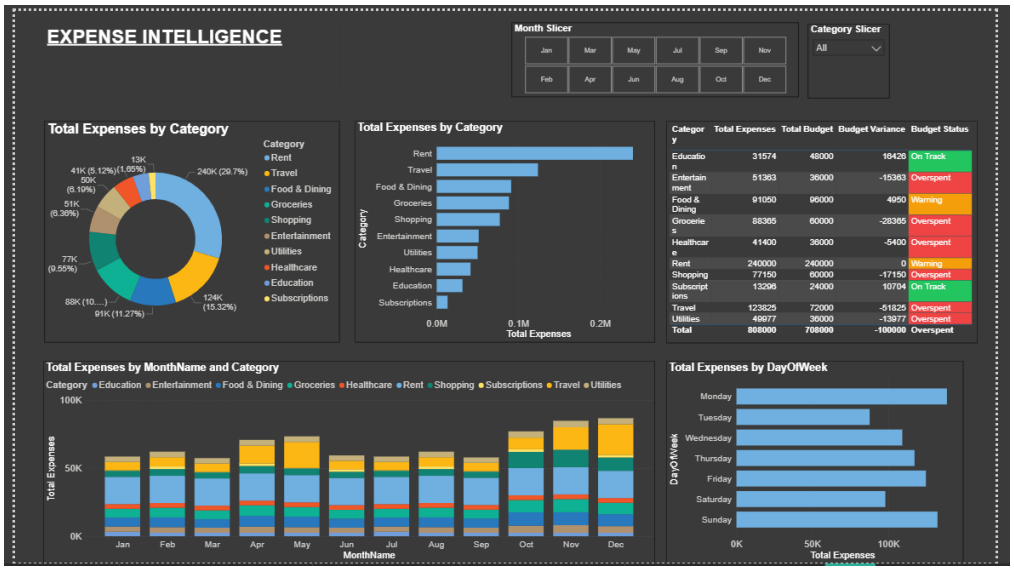

Figure 2: *Expense Intelligence showing category breakdown, budget comparison table, stacked chart, and day-of-week analysis*

Rent is the largest expense at Rs. 2,40,000, representing 29.7% of total spending. Travel follows at Rs. 1,23,825. The budget table indicates that several categories including Groceries, Shopping, and Entertainment exceeded their allocated budgets.

3.3 Dashboard 3 — Smart Insights

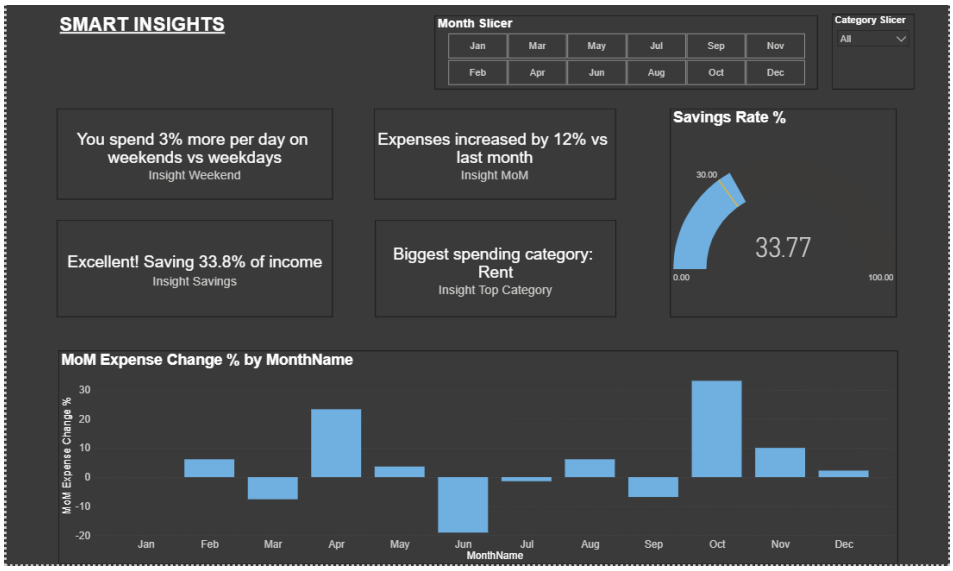

Figure 3: *Smart Insights showing auto-generated text cards, savings gauge, and month-over-month expense change chart*

The insight cards update automatically based on the selected filters. For instance, the savings card displays “Excellent! Saving 33.8% of income” and the top category card shows “Biggest spending category: Rent”. October recorded the largest month-on-month increase at approximately 30%, attributed to festive season expenditure.

3.4 Dashboard 4 — Savings Forecast

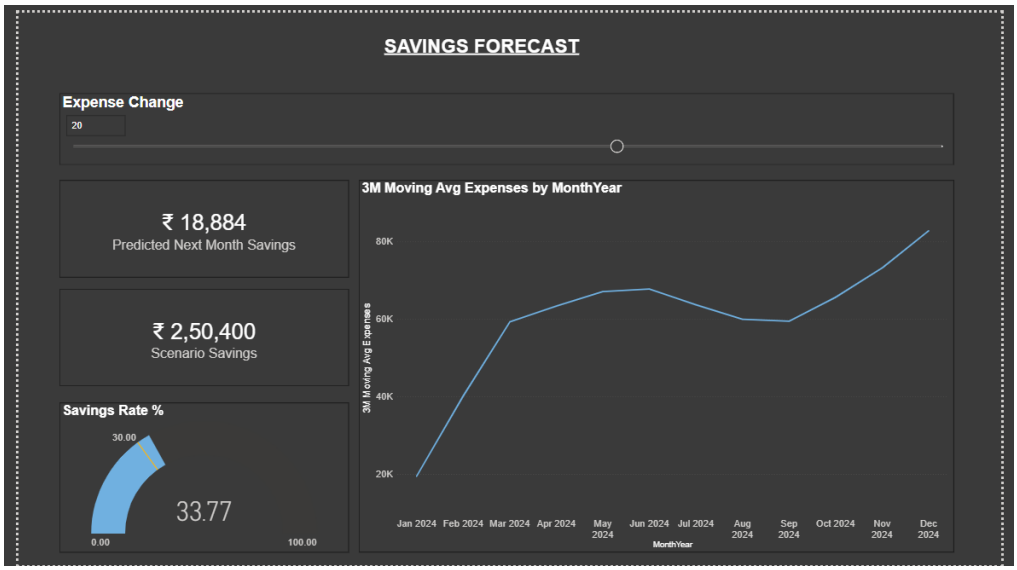

Figure 4: *Savings Forecast with What-If slider, predicted savings card, scenario savings, and 3-month moving average line chart*

With a 20% increase in expenses applied via the slider, the Scenario Savings drops to Rs. 2,50,400. The moving average chart reduces monthly fluctuations and reveals an upward trend in expenses toward the latter part of the year.

3.5 Dashboard 5 — Financial Health Score

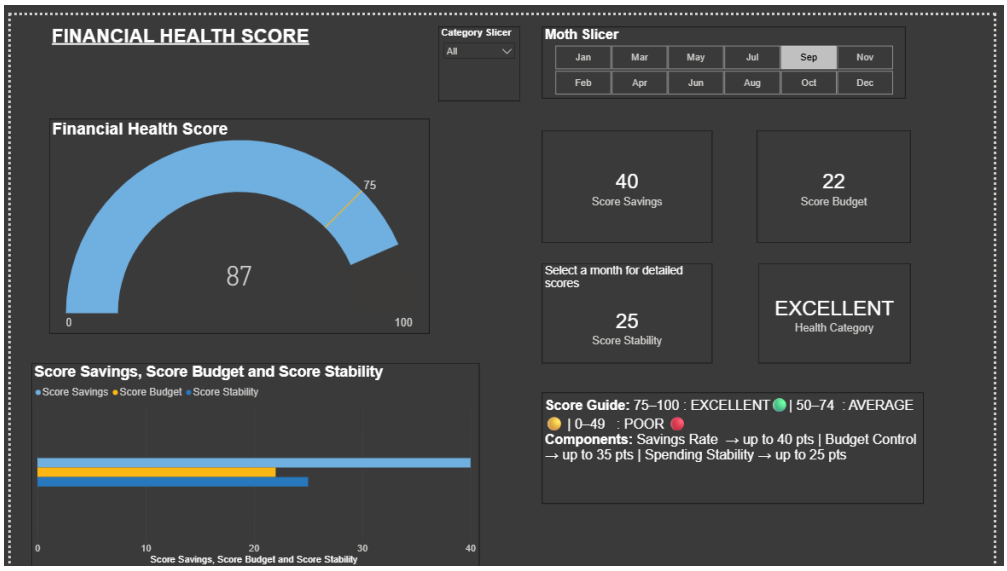

Figure 5: *Financial Health Score gauge (0 to 100) with component score cards and a breakdown bar chart*

The overall score is 53 out of 100, placing it in the AVERAGE category. The savings sub-score achieves the maximum of 40 out of 40 as the savings rate is strong. However, the budget sub-score is only 10 out of 35 as several categories exceeded their budgets.

4. Technology Stack

Technology	How it was used
Microsoft Power BI Desktop	Used to build the data model, write DAX measures, design all five dashboard pages, and configure slicers and sync settings
DAX	Used to write over 50 custom measures including KPIs, time-based calculations, text insights, and the health scoring logic
Power Query	Used to clean and transform the CSV data, enforce correct data types, and add derived columns
Star Schema	Data modelling approach used to structure the four tables and define relationships between them
CSV Files	Used as the primary data source, loaded into Power BI through the Get Data option

5. Unique Points

1. **Auto-generated insight text using DAX:** Rather than relying only on charts, this dashboard generates readable sentences directly from the data. For instance, a card may display "You spend 15% more per day on weekends than weekdays." These update automatically when filters are changed, giving the report an intelligent and responsive quality.

2. **What-If scenario planning:** A parameter slider allows users to adjust the expense percentage and immediately observe the effect on savings. This is useful for evaluating scenarios such as a 20% reduction in dining or an anticipated increase in rent. Most dashboards do not offer this kind of forward-looking capability.

3. **Financial Health Score:** Rather than presenting individual metrics in isolation, this dashboard produces a single score out of 100 based on savings rate, budget control, and spending stability. The score is divided into three components so users can identify exactly which area requires improvement.

4. **Proper Star Schema data model:** The data is structured using a Star Schema that includes a bridge table to handle the many-to-many relationship between Transactions and Budget. This mirrors the architecture used in professional business intelligence projects and is not commonly implemented in beginner-level Power BI reports.

5. **Slicers synced across all pages:** Filters applied through the Month and Category slicers are automatically reflected across all five pages using Power BI's Sync Slicers feature. Users do not need to reapply filters each time they navigate between dashboards.

6. **Colour-coded budget table:** The budget comparison table applies conditional formatting using DAX logic to highlight each category in green for On Track, orange for Warning, or red for Overspent. The colours update dynamically based on the active filter selection.

6. Future Improvements

1. **Real bank data:** The dashboard currently uses generated data. Connecting it to actual bank transaction exports or a financial API such as Plaid would make it applicable to real-world use cases.

2. **Better forecasting:** The 3-month moving average is a basic approach. Implementing a proper forecasting model using Python or R within Power BI would deliver more accurate predictions over longer time horizons.

3. **Multiple years of data:** The project covers only 2024. Including data from prior years would enable year-over-year comparisons and help identify longer-term financial trends.

4. **Mobile layout:** Power BI supports a dedicated mobile view. Designing a layout for smartphone screens would allow users to monitor their finances conveniently from a mobile device.

5. **Spending alerts:** Adding DAX-based anomaly flags for months where spending in a category deviates significantly from the average would help users detect and respond to unusual expenditure patterns.

6. **Savings goals:** A feature that allows users to define a savings target and track progress toward it over time would make the dashboard more practical for day-to-day financial planning.

7. **Multi-user access with security:** Implementing Power BI's Row-Level Security would allow the dashboard to support multiple users, with each person viewing only their own financial data within the same report.

7. Conclusion

This project covers the complete process of building a personal finance dashboard from the ground up. Beginning with a PowerShell script to generate realistic data, the project progresses into Power BI to model the data appropriately, write DAX measures, and construct five interconnected dashboard pages.

The final output is a system that allows users to track income and expenses, compare spending against the budget, simulate financial scenarios, and obtain a composite score reflecting overall financial health. The skills applied throughout, including data modelling, DAX, Power Query, and dashboard design, are directly relevant to business intelligence and data analytics roles, which is the primary intent of this project.